C. G.
DE CASIBVS

quibus integrari potest

ÆQVATIO DIFFERENTIALIS

$$ax^m dx + byx^p dx + cy^2 dx = dy.$$

OBSERVATIONES QVÆDAM.



Um Anno 1721. Venetiis inviserem Cl. Zendrinum Medicum & Mathematicum celebrem eo die quo ab urbe discedere constitueram, accidit ut Nicolai Bernoulli Ioh. Fil. mentionem faceret, eumque iam ab aliquo tempore Venetiis esse mihi affirmaret; igitur mox, ut optaveram, in amicitiam humanissimi viri perveni, eaque causa abitum meum in sextum diem distuli.

Inter colloquia quæ illo tempore habuimus, & ex quibus me multum profeciisse semper fatebor, æquationem quandam differentialem ab Ill. Riccato olim propositam mihi ostendit huiusmodi : $ax^m dx + by^2 x^p dx = dy$. ut tentarem, an quo casu illam vel separare vel integrare possem. Inveni igitur paullo post, si æquatio generalior ponatur $ay^f x^m dx + by^n x^p dx = dy$. y determinari posse per x quoties $m = \frac{fp+f-p-n}{n-1}$, nempe si c fuerit radix huius æquationis $bc^{n-1} + ac^{f-1} + \frac{p-m}{n-f} = o$. esse $y = cx^{\frac{-p-1}{n-1}}$ qui casus omnium facillimus est. Deinde datis 30. Iul. A. 1721. litteris ad Cl. Nicol. Bernoullium, tres alios ca-

ſus integrabiles æquationis $ax^m dx + by^n x^p dx = dy$. detexi, nimirum (1) ſi ponatur $n = 1, p = -1$ eſſe $y = \frac{a}{-b+m+1} + fx^b$ ubi f ſit quantitas conſtans quæcunque. (2) Si $n = 2$, $m = -3p-4$ eſſe $y = V\frac{-a}{b} x^{-2p-2} \cdot -\frac{(p+1)}{b} x^{-p-1}$. (3) Si $n = 2, m = \frac{-p-4}{3}$ eſſe $y = 1 : V\frac{-a}{b} x^{\frac{2p+2}{3}} - \frac{(p+1)}{3a} x^{\frac{p+1}{3}}$ quas integrationes ut novas placere ſibi teſtatus eſt Vir Cl. in reſponſo quod ad meas dedit; paullo poſt aliis litteris ſignificauit ſe præter caſus mihi obſervatos modum inveniſſe peculiarem ex æquatione $ax^m dx + by^2 x^p dx = dy$ alios quotlibet caſus integrabiles eruendi & pro iis æquationes algebraicas aſſignandi, rem ſimul exemplo confirmavit, poſito nempe $m = \frac{-5p-8}{3}$ eſſe

$$y = \frac{x^{-2p-2} + (\frac{p+1}{-b}) x^{\frac{-5p-5}{3}} V\frac{b}{-a} + \frac{(p+1)^2}{-3ab} x^{\frac{-4p-4}{3}}}{x^{\frac{-2p-2}{3}} V\frac{b}{-a} + \frac{(p+1)}{3a} x^{\frac{-p-1}{3}}}$$

Tandem litteris 6. Dec. A. 1721. ad me datis ſtatuit æquationem illam differentialem eſſe integrabilem ſi $m = \frac{-2np-4n+p}{2n+1}$ ubi per p intelligit numerum rationalem, per n vero numerum integrum quemcunque. Atque hæc ipſa eſt formula quam ad ſeparationem indeterminatarum huius æquationis dedit Cl. Daniel Bernoullius in Exercitation. Mathemat. p. 77. ſ. ubi per duo lemmata alternatim caſus omnes methodo perquam ingenioſa & facili deduxit.

Conſideravit autem æquationem huiusmodi $ax^n dx + uudx = bdu$. alteri illi æquivalentem ex qua ſcilicet non ſolum exponens p, ſed etiam uterque coefficiens a vel b pro lubitu in alium quemcunque transmutari poteſt.

Pona-

Ponamus exempli causa datam æquationem $cu^g du + fu^b y^2 du = dy$ reducendam esse ad hanc formam (A)... $x^m dx - y^2 dx = dy$ sumatur $u = ex^k$, $z = by$. ubi sit

$$e = -(b+1)^{\frac{2}{g+b+2}} c^{\frac{g+1}{g+b+2}} f^{\frac{-1}{g+b+2}}, \quad b = \frac{ce^{\frac{g+1}{b+1}}}{b+1}, \quad k = \frac{1}{b+1}$$

fiet $m = \frac{g-b}{b+1}$ vel, si reducenda sit eadem æquatio ad hanc formam (B)... $dx - y^2 x^m dx = dy$ sumatur $u = ex^k$, $z = by$ ubi sit $e = -(g+1)^{\frac{2}{g+b+2}} c^{\frac{g+1}{g+b+2}} f^{\frac{-1}{g+b+2}}$, $b = \frac{ce^{\frac{g+1}{g+1}}}{g+1}$, $k = \frac{1}{g+1}$ fiet $m = \frac{b-g}{g+1}$ ubi observandum est in æquatione Bernoull. $ax^{\frac{-4n}{2n+1}} dx + y^2 dx = bdy$ casus omnes exponentis $\frac{-4n}{2n-1}$, si n sit numerus affirmativus, & casus omnes $\frac{-4n}{2n+1}$ si n sit negativus reduci posse ad formulam A, omnes vero casus exponentis $\frac{-4n}{2n-1}$ si n sit negativus & omnes $\frac{-4n}{2n+1}$ si n sit affirmativus redigi ad formulam B.

Iam posito $2n-1 = p$ fiat

$$c = x^{\frac{-2n+1}{p}} + Ap^{\frac{(n-0)}{n+0}} x^{\frac{-2n-0}{p}} + Bp^{\frac{(n-1)}{2n-1}} x^{\frac{-2n-1}{p}} + Cp^{\frac{(n-2)}{3n-3}} x^{\frac{-2n-2}{p}}$$
$$+ Dp^{\frac{(n-3)}{4n-6}} x^{\frac{-2n-3}{p}} + Ep^{\frac{(n-4)}{5n-10}} x^{\frac{-2n-4}{p}} + Fp^{\frac{(n-5)}{6n-15}} x^{\frac{-2n-5}{p}} + \&c.$$

$$f = 1 + Ap^{\frac{(n-1)}{n-1}} x^{\frac{-1}{p}} + Bp^{\frac{(n-2)}{2n-3}} x^{\frac{-2}{p}} + Cp^{\frac{(n-3)}{3n-6}} x^{\frac{-3}{p}}$$
$$+ Dp^{\frac{(n-4)}{4n-10}} x^{\frac{-4}{p}} + Ep^{\frac{(n-5)}{5n-15}} x^{\frac{-5}{p}} + Fp^{\frac{(n-6)}{6n-21}} x^{\frac{-6}{p}} + \&c.$$

leges quibus progrediuntur harum serierum termini perspicuæ sunt, crescunt enim omnes numeri in coefficientibus ordine naturali præter ultimos denominatorum qui servant progressionem triangularium 1, 3, 6, 10, &c. litteræ maiores A, B, C, &c. recepto more indicant coefficientes quæ proxime antecedunt.

His præmissis considero æquationem generalem

$cu^g du + fu^b y^2 du = dy$ ut iam reductam ad alterutram formularum A vel B & infinitorum illorum casuum integrationes uno theoremate complector: Æquationis differ. $x^{\frac{-(n+2n)}{2n-1}} dx - x^{\frac{(-2n+2n)}{2n-1}} y^2 dx = dy$ (ubi n sit numerus integer affirmativus) integralis est $y = \alpha \rightleftharpoons \beta$. Signum $\rightleftharpoons$ denotat divisionem ambiguam, si enim in formula æquationis diff. ex signis $\mp \pm$, datum fuerit superius (qui est casus formulæ A) sumendus est in integrali divisor quem indicat angulus a linea superiore ad horizontalem factus nempe β; si inferius signum in æquatione diff. datum fuerit, sumatur in integrali divisor quem inferior linea ad horizontalem ascendens designat ut fiat $y = \frac{\beta}{\alpha}$.

Ne quis vero in usu huius canonis hæsitet, exemplis nonnullis eundem illustrabimus quibus intellectis ulteriores casus non erunt difficiles.

I. Sit integranda æquatio $x^{-4} dx - y^2 dx = dy$ quæ pertinet ad formulam A, fiet $n = 1$. $p = 1$. $\alpha = x^{-1} + x^{-2}$, $\beta = 1$. Ergo integralis quæsita $y = \frac{\alpha}{\beta} = x^{-1} + x^{-2}$ unde sequitur æquationis $dx - y^2 x^{-4} dx = dy$ integralem esse $y = \frac{\beta}{\alpha} = \frac{1}{x^{-1} + x^{-2}}$.

II. Sit integranda æquatio $x^{\frac{-8}{3}} dx - y^2 dx = dy$, fiet $n = 2, p = 3$, $y = \frac{\alpha}{\beta} = \frac{x^{\frac{-3}{3}} + 3x^{\frac{-4}{3}} + 3x^{\frac{-5}{3}}}{1 + 3x^{\frac{-1}{3}}}$, unde sequitur æquationis $dx - y^2 x^{\frac{-8}{3}} dx = dy$ integralem esse $y = \frac{\beta}{\alpha} =$

$$\frac{\beta}{\alpha} = \frac{1+3x^{\frac{-1}{3}}}{x^{\frac{-3}{3}}+3x^{\frac{-4}{3}}+3x^{\frac{-5}{3}}}$$

III. Eodem modo æquationis $x^{\frac{-(6+6)}{5}}dx - y^2x^{\frac{-(6+6)}{5}}dx = dy$ integralis invenietur

$$y = \alpha \rightleftharpoons \beta = \left(x^{\frac{-5}{5}} + 5x^{\frac{-6}{5}} + 10x^{\frac{-7}{5}} + \frac{25}{3}x^{\frac{-8}{5}}\right) \rightleftharpoons \left(1+5x^{\frac{-1}{5}}+\frac{25}{3}x^{\frac{-2}{5}}\right)$$

IV. Æquationis $x^{\frac{-(8+8)}{7}}dx - y^2x^{\frac{(8+8)}{7}}dx = dy$. integralis est

$$y = \left(x^{\frac{-7}{7}} + 7x^{\frac{-8}{7}} + 21x^{\frac{-9}{7}} + \frac{98}{3}x^{\frac{-10}{7}} + \frac{343}{15}x^{\frac{-11}{7}}\right) \rightleftharpoons \left(1+7x^{\frac{-1}{7}}+\frac{98}{5}x^{\frac{-2}{7}}+\frac{343}{15}x^{\frac{-3}{7}}\right).$$

Quodsi iam progrediamur ad æquationem differentialem uno termino ampliorem $ax^m dx + byx^p dx + cy^2 dx = dy$ (ex qua tamen duæ coefficientium a, b, c, pro lubitu in alias transmutari possunt, ut supra indicatum fuit) similiter varii se offerunt casus integrabiles quorum nonnullos explicabimus.

I. Si $ax^{\frac{4n+2b+4}{-2n+1}}dx + byx^{-1}dx + cy^2dx = dy$. æquatio erit integrabilis dato n numero quocunque integro.

Demonstr.

Ponatur $y = \frac{-(b+1)}{c}x^{-1} + x^{-b-2}u$ transformabitur æquatio data in hanc

$$ax^{\frac{4n+2b+4}{-2n+1}}dx-(\tfrac{b^2+b}{c})x^{-2}\,dx+bx^{-b-3}\,udx$$
$$+(b^2+2b+1)x^{-2}\,dx-(2b+2)x^{-b-3}\,udx+cx^{-2b-4}\,u^2dx$$
$$=(\tfrac{b+1}{c})x^{-2}\,dx+(b+2)x^{-b-3}\,udx+x^{-b-2}$$

sive destructis terminis æqualibus, $ax^{\frac{4n+2b+4}{-2n+1}}dx+cx^{-2b-4}u^2dx=x^{-b-2}\,du$, multiplicentur omnia per x^{+b+2} fit $ax^{\frac{-2nb+b+2}{-2n+1}}dx+cx^{-b-2}\,u^2dx=du$, tandem ponatur $x=v^{\frac{-1}{b+1}}$. fiet $\frac{-a}{b+1}v^{\frac{4n}{-2n+1}}dv,\frac{-c}{b+1}u^2dv=du$, quam constat esse integrabilem si n fit numerus quicunque integer.

Potest etiam sic demonstrari: sit $ax^{\frac{-4f+2b}{2f+1}}dx+byx^{-1}dx+cy^2dx=dy$, (quæ æquatio non differt a priori, si ponatur $n=f+1$) fiat $y=x^bu$ mutabitur æquatio in hanc $ax^{\frac{-4f+2b}{2f+1}}dx+cx^{2b}u^2dx=x^bdu$, divisis omnibus per x^b fit $ax^{\frac{-4f+b-2fb}{2f+1}}dx+cx^bu^2dx=du$. Ponatur $x=z^{\frac{1}{b+1}}$. fict $\frac{a}{b+1}z^{\frac{-4f}{2f+1}}dz+\frac{c}{b+1}u^2dz=du$, quam constat esse integrabilem si f sit numerus quicunque integer.

Exempl. 1.

Si $m=2b$ (cui respondet $n=1$) hoc est, si æquatio data sit $ax^{2b}dx+byx^{-1}dx+cy^2dx=dy$

erit $y=(-ax)^{\frac{1}{2}}x^b$.

Exempl. 2.

Si $m=\frac{2b-4}{3}$, (cui respondet $n=-2$) hoc est si æqua-

quatio data ſit $ax^{\frac{2b-4}{3}}dx+byx^{-1}dx+cy^{2}dx=dy$.

erit $y=\frac{1}{(-c:a)^{\frac{1}{2}}x^{\frac{-b+2}{3}}-\frac{(b+1)}{3a}x^{\frac{-2b+1}{3}}}$

Excempl. 3.

Si $f=1$ hoc eſt, ſi æquatio data ſit $ax^{-2b-4}dx+byx^{-1}dx+cy^{2}dx=dy$, erit $y=(-a:c)^{\frac{1}{2}}x^{-b-2}-\frac{(b+1)}{c}x^{-1}$. atque ſic in ſimilibus exemplis determinari poterit valor y per x & conſtantes; ſingularis autem caſus eſt ubi m non determinatur per coefficientes datas, ſcilicet ſi fuerit $ax^{-2}dx+byx^{-1}dx+cy^{2}dx=dy$, ubi fit

$y=\left(\frac{-(b+1)+\sqrt{(b+1)^{2}-4ac}}{2c}\right)x^{-1}$. Gabriel Manfredius in tract. de conſtructione æquationum differ. primi gradus p. 167. art. 105. cum incidiſſet in æquationem $nx^{2}dx-ny^{2}dx+x^{2}dy=xydx$. *hæc*, inquit, *æquatio non apparet quomodo conſtruibilis ſit, neque enim videmus quomodo illam integremus, nec quomodo indeterminatas ab invicem ſeparemus.* Sed integrari poteſt per formulam noſtram hoc modo: dividatur æquatio per x^{2}, fiet $ndx-x^{-1}ydx-nx^{-2}y^{2}dx=dy$, poſitoque $x=z^{-1}$ erit $-nz^{-2}dz+z^{-1}ydz+ny^{2}dz=dy$. Ergo $y=\left(\frac{-1+\sqrt{n^{2}+1}}{n}\right)z^{-1}$.

II.

Si æquatio A... $ax^{p-1}dx+byx^{p}dx+cy^{2}dx=dy$ integrari poteſt, erit etiam æquatio

B... $\frac{(abp+a^{2}c)}{pp}x^{2p}+\frac{(2ac+bp)}{p}x^{p}udx+cu^{2}dx=du$

integrabilis.

Demonſtr.

Demonſtratio.

Ponatur $y=\frac{a}{p}x^p+u$, fiet

$$\begin{aligned} ax^{p-1}dx &\quad - \quad - \quad - \quad ax^{p-1}dx \\ byx^p dx &= \frac{ab}{p}x^{2p}dx+bux^p dx \\ cy^2 dx &= \frac{a^2c}{p^2}x^{2p}dx+\frac{2ac}{p}ux^p dx+cu^2 dx \\ = dy &= du+ax^{p-1}dx^{p^2} \end{aligned}$$

quæ eſt ipſa æquatio B. ſi pro u ponatur y, atque ſimiliter, ſi aliqua æquatio integrabilis revocari poſſit ad B demonſtratum eſt æquationem A eſſe integrabilem.

Exempl. 1.

Cum æquatio $ax^{p-1}dx+byx^p dx+\frac{b}{a}y^2 dx=dy$, ſit integrabilis $\left(\text{fit enim } y=\frac{a}{b}x^{-1}.\right)$ erit etiam $\left(\frac{ab+abp}{pp}\right)x^{2p}dx+\frac{(2b+bp)}{p}yx^p dx+\frac{b}{a}y^2 dx=dy$ integrab. quæ æquatio revocari poteſt ad hanc formam

$$ax^{\frac{(b^2-4ac)\pm b\sqrt{b^2-4ac}}{ac}}dx+byx^{\frac{(b^2-4ac)\pm b\sqrt{b^2-4ac}}{2ac}}dx+cy^2 dxdy.$$

unde eruitur $y^{\frac{-b\pm\sqrt{b^2-4ac}}{2c}}x^{\frac{(b^2-4ac)\pm b\sqrt{b^2-4ac}}{2a}}=\frac{c}{r}x^{-1}$.

Exempl. 2.

Quoniam æquatio $ax^{\frac{-2b-ac}{b}}dx+byx^{\frac{-b-ac}{b}}dx+cy^2 dx=dy$ eſt integrabilis, $\left(\text{fit enim } y=\frac{-b}{c}x^{\frac{-b-ac}{b}}-\frac{1}{c}x^{-1}\right)$ erit

erit etiam $\frac{-ab^3}{(b+ac)^2} x^{\frac{-2b-2ac}{b}} dx + \frac{abc-b^2}{(-b+ac)} x^{\frac{-b-ac}{b}} u dx$

$+cu^2 dx = du.$

integrabilis & $u = -\frac{-bb}{bc+ac^2} x^{\frac{-b-ac}{b}} - \frac{1}{c} x^{-1}$.

Exempl. 3.

Quoniam ex æquatione $ax^{\frac{ac-2b}{b}} dx + byx^{\frac{ac-b}{b}} dx$

$+cy^2 dx = dy$, fit $y = \frac{1}{\frac{ac-b}{ab} x^{\frac{-ac+b}{b}} - \frac{b}{a} x}$

Erit etiam $\frac{(2a^2c-ab)b^2}{(ac-b)^2} x^{\frac{2ac-2b}{b}} dx + \frac{(3abc-b)^2}{ac-b} x^{\frac{ac-b}{b}} u dx$

$+cu^2 dx = du$,

integrabilis & $u = \frac{b^2}{ac-b} x^{\frac{ac}{b}}$.

III.

In quibuscunque casibus integrari potest

A — — $ax^m dx + byx^p dx + cy^2 dx = dy.$

poterit etiam integrari

B — — $\frac{-c}{m+1} v^{\frac{-m}{m+1}} dv \frac{-bu}{m+1} v^{\frac{p-m}{m+1}} dv = \frac{a}{m+1} u^2 dv = du.$

Demonstr.

Ponatur $y = u^{-1}$ mutabitur æquatio A in

$ax^m dx + bu^{-1} x^p dx + cu^{-2} dx = -u^{-2} du.$

Hæc multiplicetur per $-u^{-2}$ fiet $-cdx - bux^p dx - ax^m$

$u^2 dx = du$. positoque $x = v^{\frac{1}{m+1}}$ prodibit

B — — $\frac{-c}{m+1} v^{\frac{-m}{m+1}} dv + \frac{-b}{m+1} uv^{\frac{p-m}{m+1}} dv \frac{-a}{m+1} u^2 dv = du.$

Exempl. 1.

Quoniam æquatio $ax^{\frac{-2b-ac}{b}}dx+byx^{\frac{-b-ac}{b}}dx+cy^2dx=dy$ eſt integrabilis, ut antea dictum, poterit etiam integrari $\frac{bc}{ac+b}v^{\frac{ac+2b}{-ac-b}}dv+\frac{b}{ac+b}^2v^{\frac{b}{-ac-b}}udv+\frac{ab}{ac+b}u^2dv=du$. quæ reduci poteſt ad

$ax^{\frac{-ac-2}{ac+1}}dx+byx^{\frac{-1}{ac+1}}dx+cy^2dx=dy$.

Exempl. 2.

Quoniam æquatio $ax^{\frac{ac-2b}{b}}dx+byx^{\frac{ac-b}{b}}dx+cy^2dx=dy$ eſt integrabilis (vid. ſup. Ex. 3.) erit etiam

$\frac{-cb}{ac-b}x^{\frac{2b-ac}{ac-b}}dx-\frac{b}{ac-b}^2ux^{\frac{b}{ac-b}}dx-\frac{ab}{ac-b}u^2dx=du$.

quæ revocari poteſt ad hanc formam

$ax^{\frac{2-ac}{ac-1}}dx+byx^{\frac{1}{ac-1}}dx+cy^2dx=dy$ integrabilis.

Exempl. 3.

Quoniam $ax^{-2}dx+byx^{\frac{-1\pm\sqrt{1-4ac}}{2}}dx+cy^2dx=dy$ eſt integrabilis, fit enim $y=\left(\frac{-1\pm\sqrt{1-4ac}}{2c}\right)x^{-1}$

$\frac{-b}{c}x^{\frac{-1\pm\sqrt{1-4ac}}{2}}$ erit etiam integrabilis.

$ex^{-2}dx+bux^{\frac{-3\pm\sqrt{1-4ac}}{2}}dx+ay^2dx=dy$

& fiet $y=1:\left(\frac{(-1\pm\sqrt{1-4ac}}{2a}x-\frac{b}{a}x^{\frac{1\pm\sqrt{1-4ac}}{2}}\right)$

IV.

Si in æquatione generali ponatur $c=0$ æquatio erit integrabilis dato $\frac{m-p}{p+1}$ numero integro affirmativo, poteſt autem hæc æquatio ad ſimpliciores terminos reduci, ſi po-

si ponatur $x = z^{\frac{1}{p+1}}$, fiet enim $az^{\frac{m-p}{p+1}}dz + bydz = dy$. Ergo $ax^n dx + bydx = dy$ erit integrabilis si n sit numerus integer positivus.

Exempla.

Si $n = 1$ erit $y = \frac{-a}{b}x - \frac{a}{bb}$

Si $n = 2$ erit $y = \frac{-a}{b}x^2 - \frac{2ax}{bb} - \frac{2a}{b^3}$

Si $n = 3$ erit $y = \frac{-a}{b}x^3 - \frac{3a}{bb}x^2 - \frac{6a}{b^3}x - \frac{6a}{b^4}$

& generatim dato n numero quocunque erit $y =$ seriei infinitæ

$$\frac{-a}{b}x^n - \frac{an}{b^2}x^{n-1} - \frac{an(n-1)}{b^3}x^{n-2} - \frac{an(n-1)(n-2)}{b^4}x^{n-3} - \&c.$$

quæ toties abrumpitur quoties n est numerus integer positivus.

Si vero ponamus

$y = \alpha x^e u + \beta x^f u^2 + \gamma x^g u^3 + \delta x^h u^4 + \varepsilon x^k u^5 + \&c.$

transmutabitur æquatio $ax^m dx + bydx = dy$ in

$ax^m dx + b\alpha x^e u dx + b\beta x^f u^2 dx + \&c.$

$= (\alpha x^e + 2\beta x^f u + 3\gamma x^g u^2 + \&c.)du +$

$(\alpha x^{e-1} u + \beta f x^{f-1} u^2 + \gamma g x^{g-1} u^3 + \&c.)\, dx$. quæ similiter erit integrabilis si m sit numerus integer positivus ; ad quam formam reduci possunt plurimi casus particulares quorum integratio sine hoc theoremate non facile succederet.

Exempl. 1.

Sit data æquatio $x^3 dx + (2x-1)udx - 2u^2 dx = (x-2u)du$ hoc est $\alpha = a = e = 1$. $f = \gamma = \delta = \&c. = 0$. $\beta = -1$. $b = 2$. $m = 3$. habebitur $u = x \pm \sqrt{2x^3 + 4x^2 + 3x + 3}$

Exempl. 2.

Sit æquatio data

$$ax^2dx+\tfrac{2}{5}uxdx-\tfrac{b}{5}ux^2dx+u^3dx-bu^3xdx-bu^5dx$$
$$=-\tfrac{1}{5}x^2du-3xu^2du-5u^4du.$$

erit $\alpha=-\frac{1}{5}$. $\beta=0$. $\gamma=-1$. $\delta=0$. $\varepsilon=-1$. & reliqui coefficientes $=0$. $e=2$. $g=-1$. $k=0$. $m=2$. quam ob rem ipsa æquatio dextre tractata perducetur ad

$$u=\left(\frac{(-ab^2x^2-2abx-2a)+\sqrt{(ab^2x^2+2abx+2a)^2+x^5}}{2b^3}\right)^{\frac{1}{5}}$$
$$+\left(\frac{(+ab^2x^2+2abx+2a)-\sqrt{(ab^2x^2+2abx+2a)^2+x^5}}{2b^3}\right)^{\frac{1}{5}}$$

V.

Æquatio $ax^{\frac{-2ac-b}{b}}dx+bx^{\frac{-2ac}{b}}udx+cu^2dx=du$ est integrabilis, si $\frac{\mp ac}{2ac-b}$ sit numerus integer.

Demonstr.

Si ponatur $u=y-\frac{b}{2c}x^{\frac{-2ac}{b}}$ fiet

$$\left\{\begin{array}{l} ax^{\frac{-2ac-b}{b}}dx \\ +bx^{\frac{-2ac}{b}}udx=by^{\frac{-2ac}{b}}dx-\frac{b^2}{2c}x^{\frac{-4ac}{b}}dx. \\ +cu^2dx=-by^{\frac{-2ac}{b}}dx+\frac{b^2}{4c}x^{\frac{-4ac}{b}}dx+cy^2dx \end{array}\right.$$

$$=du=ax^{\frac{-2ac-b}{b}}dx \quad - - - - - - - +dy.$$

hoc est, omissis terminis qui sese destruunt,

$$-\tfrac{b^2}{4c}x^{\frac{-4ac}{b}}dx+cy^2dx=dy.$$

quam constat esse integrabilem si $\frac{\mp ac}{2ac-b}$ sit numerus integer.

VI.

VI.

Æquatio $ax^{2e}dx+bux^{e}dx+\frac{b^2}{4a}u^2dx=du$ eſt integrabilis ſi $\frac{\pm e\mp 1}{2e+2}$ ſit numerus integer.

Demonſtr.

Si ponatur $u=y-\frac{2a}{b}x^{e}$ fiet

$$\left.\begin{array}{l}\int ax^{2e}dx\\ +bux^{e}dx=-2ax^{2e}dx+byx^{e}dx\\ +\frac{b^2}{4a}u^2dx=ax^{2e}dx-byx^{e}dx+\frac{b^2}{4a}y^2dx\end{array}\right\}$$

$$=du=-\frac{2ae}{b}x^{e-1}dx \quad - \quad - \quad - \quad - \quad - \quad +dy.$$

ſeu omiſſis terminis qui ſeſe deſtruunt $\frac{2ae}{b}x^{e-1}dx+\frac{b^2}{4a}y^2dx=dy$ quam conſtat eſſe integrabilem ſi $\frac{\pm e\mp 1}{2e+2}$ ſit numerus integer.